

Course Outline

This outline provides an overview of the course and assignments by week. Please remember to follow the assignment due dates. Each course module runs for a period of seven (7) days, i.e., one week. Due dates for readings and other assignments are referred to by the day of the module week in which they are due. All problem sets are due the evening before the next module begins. Thus, for example, the first homework is due by midnight on Tuesday the 3rd of September.

Note that our 'Boot-Camp' problems in the first two Modules are mainly to help you get started. These will be lightly graded.

Module	Dates	Topics	Assignments
Module 1	Wed 8/27/25– Tue 9/2/25	'Boot-Camp': part I Introducing Measure Theory	Reading Assignment (in Capinski and Copp), First Discussion Contribution and 1st Problem Set
Module 2	Wed 9/3/25– Tue 9/9/25	'Boot-Camp': part II Measures and Integrals	Reading Assignment (in Capinski and Copp), 2nd Discussion Contribution and 2nd Problem Set
Module 3	Wed 9/10/25– Tue 9/16/25	Introduction to Oksendal and SDEs	Reading Assignments (Oksendal, and all reading is that text from this point on: Chapters 1 and 2) Third Discussion Contribution 3 rd Problem Set
Module 4	Wed 9/17/25– Tue 9/23/25	Ito Integrals	Reading Assignments (Chapter 3) Discussion Contribution 4 nd Problem Set
Module 5	Wed 9/24/25– Tue 9/30/25	Ito Formulas and Martingales	Reading Assignments (Chapter 4) Discussion Contribution 5 rd Problem Set, Project Topic Due
Module 6	Wed 10/1/25– Tue 10/7/25	Stochastic Differential Equations	Reading Assignments (Chapters 5) Discussion Contribution 6 th Problem Set
Module 7	Wed 10/8/25– Tue 10/14/25	Filtering Problems	Reading Assignments (Chapters 6) Discussion Contribution 7 th Problem Set, Quiz-1
Module 8	Wed 10/15/25– Tue 10/21/25	Diffusions	Reading Assignments (Chapter 7) Discussion Contribution 8 th Problem Set Outline/Bibliography for Project Due
Module 9	Wed 10/22/25– Tue 10/28/25	Feynman-Kac and Backward Kolmogorov Equations	Reading Assignments (Chapter 8) Discussion Contribution 9 th Problem Set Pseudo-code of Project Due
Module 10	Wed 10/29/25– Tue 11/4/25	Dirichlet and Poisson Boundary Value Problems	Reading Assignments (Chapter 9) Discussion Contribution 10 th Problem Set
Module 11	Wed 11/5/25– Tue 11/11/25	Optimal Stopping	Reading Assignment (Chapter 10) Discussion Contribution 11 th Problem Set, Quiz-2

Module 12	Wed 11/12/25– Tue 11/18/25	Stochastic Control	Reading Assignment (Chapter 11) Discussion Contribution 12 th Problem Set Fifty-percent of Project Due
Module 13	Wed 11/19/25– Tue 11/25/25	HJB Equations	Reading Assignment (Chapter 11) Discussion Contribution 13 th Problem Set Quiz-3
Thanksgiving Break	Wed 11/26/25 - Tue 12/2/25	(no assignments)	(no homework due)
Module 14	Wed 12/3/25– Tue 12/9/25	Black-Scholes, Options and Markets	Reading Assignment (Chapter 12) Discussion Contribution Final Project Due